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A study on stomatal pattern in some of the members of Asteraceae of Ranchi district

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Abstract : A study on epidermal peels of some of the members of the Family Asteraceae of Ranchi District revealed that Hypoamphistomatic nature is common in the members. The size of stomata varied from one species to another as well within the species. The stomatal index, stomatal frequencies, length and breadth of stomata and l/b ratio of stomata were calculated for both the surfaces of these species. These studies were mainly based on tracing technique.

Keywords : Stomatal pattern, Leaf peels, Asteraceae, Ranchi District

INTRODUCTION

The taxonomic value of leaf epidermal characters is well documented by different researchers (Inamdar[1], Hayat[2], Hickey[3], Sen[5], Nautiyal[6]). Met-calfe and Chalk gave a sparse description of the gen-eral anatomy of the family *Asteraceae*. Hickey worked on various aspects of leaf architectural design and venation pattern of present and ancient leaves. The available information, the report on the family *Asteraceae* in India is very scanty. The present study is therefore aimed at iden-tifying, describing and documenting the leaf epidermal characters that are diagnostic and taxonomically important in the family *Asteraceae* occurring in Ranchi.

MATERIALAND METHOD

Fresh leaves were collected, epidermal peels were taken out and slides were prepared following the method used by Foster and modified by Hickey.

A new technique was developed to make the impressions of leaves as human finger printing. It gave a better result and better understanding of leaf surfaces in short period of time. It is better in the sense as less time is taken for making slides, exact positioning of the epidermal features, independent of texture of leaf, easy to measure the length and breadth of any epidermal structure like stomata and trichomes and easy to maintain slides. To make a slide the surface of the leaf was washed and dried and cut into the pieces of 2cm x 2 cm size. The dried slide was first labeled with marker by writing the scientific name of the plant, date of making slide, upper (U) or lower (L) sides of leaf and other necessary information's required in future. On the other side of the slide a drop of super glue (Loctite 495) was put above the U mark (on the opposite side of the slide having U mark) and the upper surface of the cut leaf piece was pasted. This was similarly repeated for the cut pieces of lower surface of the leaf. Slide was left for drying for 2-5 minutes and then the pasted leaf pieces were pulled out with precautions with the help of forceps. The time needed for fixation depend upon the type of glue used and the surface of the leaf. If necessary a clip press may be used to keep leaf pieces flat. The clip was removed after two minutes to five minutes.

This technique is suitable for any kind of leafy material except the leaves with excessive hairs/ trichomes on the surface. Slides of those leaves can be easily

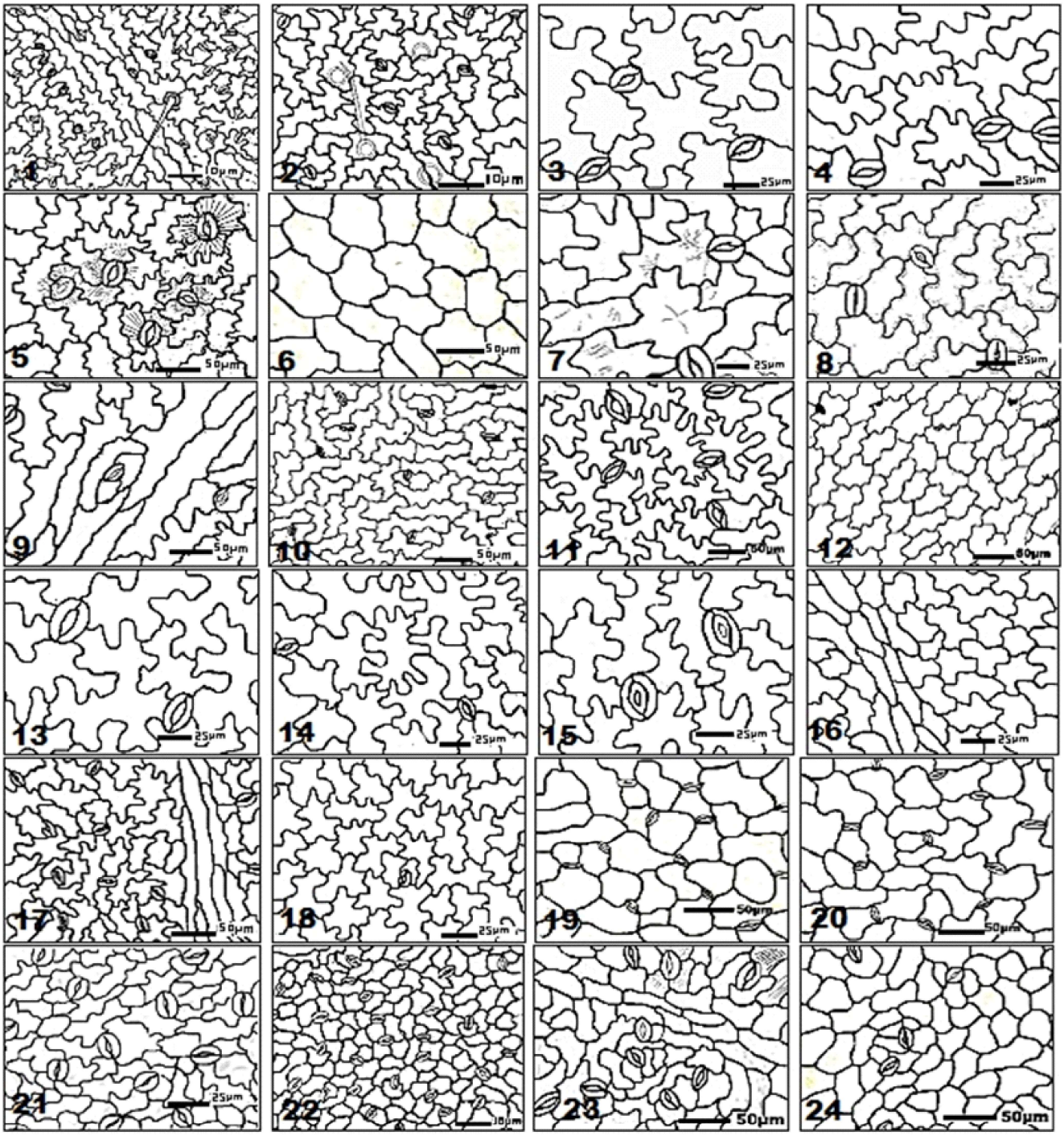
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prepared having very thin texture like ***Cythocline*** having very thin texture or pinnately compound leaves of ***Glossocardia*** having very thin leaf lobes. Slides made by above technique helped in easy study of leaf texture, stomata shape, stomata size, arrangement of cells, cell margins, shape and size of trichomes, etc. Stomatal index can be easily counted and stomata of different plants can be easily compared.

RESULT AND DISCUSSION

It was observed that many of the studied plants were amphistomatic in stomatal distribution where Hypoamphistomatic nature was more common than Epiamphistomatic. Plants like ***Blumea sinuta*** (Lour.) Merr., ***Chrysanthemum coronarium*** L., ***Parthenium hysterophorus*** L., ***Spilanthes calva*** DC. were Epiamphistomatic while ***Cyathocline purpurea*** (Buch.-Ham.ex D. Don) Kuntze, ***Emilia parnanthoides*** DC., ***Dahlia imperialis*** Ortgies, ***Galinsoga quadriradiata*** Ruiz & Pav, ***Launea procumbens*** (Roxb.) Ramayya & Rajagopal, ***Solidago canadensis*** L. and ***Sonchus wightianus*** DC. were Hypoamphistomatic. ***Calyptocarpus vialis*** Less., ***Chrysanthemum indicum*** L. and***Eupatorium odoratum*** L., were the only studied members having stomata only on lower surface of the leaf, i.e. Hypostomatic. The stomatal frequency of the lower epidermis of the leaves of different members varied from 16.18 in ***Chrysanthemum indicum*** L. to 33.86 in ***Blumea sinuta*** (Lour.) Merr., while in upper epidermis, this range was 2.41 in ***Dahlia imperialis*** Ortgies to 34.25 in ***Chrysanthemum coronarium*** L. The stomatal index in lower surface of the leaves varied from 13.91 in ***Parthenium hysterophorus*** L. to 25.28 in ***Blumea sinuta*** (Lour.) Merr. The size of stomata varied not only from one plant to another but also with in a plant. The maximum range 44.39 µm x 25.37 µm in lower epidermis was observed in the ***Dahlia imperialis*** Ortgies while lowest range 16.58 µm x 7.61 µm was in ***Sonchus wightianus*** DC. The length of stomata varied from 41.22 µm to 47.56 µm while the breadth varied from 19.02 µm to 28.54 µm in ***Dahlia imperialis*** Ortgies. There was a similar variation observed in the shape and size of the stomata present on the upper epidermis. The maximum range 50.24 µm x 21.86 µm in upper epidermis was observed in the ***Dahlia imperialis*** Ortgies while lowest range 5.14 µm x 3.75 µm was in ***Eupatorium odoratum*** L. The length of stomata varied from 44.39 µm to 57.07 µm while the breadth varied from 21.00 µm to 22.20 µm in ***Dahlia imperialis*** Ortgies. The length and breadth ratio of stomata was also found out for many species. It was observed that this range varied from 1.25 in the ***Solidago canadensis*** L. in upper epidermis to 2.27 in upper epidermis of the ***Launea procumbens*** (Roxb.) Ramayya & Rajagopal. and from 1.32 in ***Eupatorium odoratum*** L. to 2.29 in ***Launea procumbens*** (Roxb.) Ramayya & Rajagopal. This range may indicate the opening capacity of the stomata directly related to diffusion of gases.

EPIDERMAL PEELS OF LOWER (LE) AND UPPER EPIDERMIS (UE) OF LEAVES OF DIFFERENT SPECIES OF ASTERACEAE (Tracings of microphotographs)



1. ***Blumea sinuata***(Lour.) Merr. **(1. LE 2. UE)**
2. ***Chrysanthemum coronarium*** L. **(3. LE 4. UE)**
3. ***Chrysanthemum indicum*** L. **(5. LE 6. UE)**
4. ***Cyathocline purpurea*** (Buch.-Ham. ex D. **(7. LE 8. UE)**
5. ***Emilia prenanthoides*** DC. **(9. LE 10. UE)**
6. ***Dahlia imperialis*** Ortgies **(11. LE 12. UE)**
7. ***Galinsoga quadriradiata***Ruiz & Pav. **(13. LE 14. UE)**
8. ***Eupatorium odoratum*** L. **(15. LE 16. UE)**
9. ***Calyptocarpus vialis***Less. **(17. LE 18. UE)**
10. ***Launea procumbens***(Roxb.) Ramayya & Rajagopal **(19. LE 20. UE)**
11. ***Parthenium hysterophorus*** L. **(21. LE 22. UE)**
12. ***Solidago canadensis*** L. **(23. LE 24. UE)**

LE= lower epidermis **Up**= upper epidermis

The cell wall outline varied from one species to another. The undulating or wavy nature was the most common feature in the family. It was highly undulate in both upper and lower surface of leaves of ***Galinsoga quadriradiata*** Ruiz & Pav, ***Spilanthes calva*** DC., ***Crassocephalum crepidioides*** (Berth.) S. Moore and ***Calyptocarpus vialis*** Less. This undulate nature was present in the upper epidermis of ***Emilia parnanthoides*** DC., ***Chrysanthemum coronarium*** L., ***Cyathocline purpurea*** (Buch.-Ham.ex D. Don) Kuntze and on lower surface of the leaves of ***Eupatorium odoratum*** L. and ***Chrysanthemum indicum*** L. but not on the other surface.

SN.	Name of the plant	Type of leaf	Stomatal features of lower leaf					Stomatal features of upper leaf				
			Length	Breadth	L/B ratio	Stomatal Index	stomatal frquency	Length	Breadth	L/B ratio	Stomatal Index	Stomatal frquency
1	***Blumea sinuata (Lour.) Merr.***	Epiamphistomatic	28.33	19.17	1.48	25.28	33.86	14.36	9.04	1.59	25.46	34.17
2	***Calyptocarpus vialis*** Less.	Hypostomatic	20.20	15.07	1.34	17.72	21.60	13.30	8.73	1.52	2.25	2.37
3	***Chrysanthemum coronarium*** L.	Epiamphistomatic	21.60	15.70	1.38	19.49	24.47	20.68	14.69	1.41	25.31	34.25
4	***Chrysanthemum indicum*** L.	Hypostomatic	38.64	25.52	1.51	14.18	16.18	35.02	22.45	1.56	<1	<1
5	***Cyathocline purpurea*** (Buch.-Ham. ex D. Don) Kuntze .	Hypoamphistomatic	27.15	18.75	1.45	17.78	21.62	31.90	21.13	1.51	11.29	12.73
6	***Emilia prenanthoides*** DC.	Hypoamphistomatic	30.73	17.12	1.79	25.23	33.78	36.37	17.16	2.12	10.94	12.38
7	***Dahlia imperialis*** Ortgies	Hypoamphistomatic	44.39	25.37	1.75	20.39	26.40	50.24	21.86	2.30	2.34	2.41
8	***Eupatorium odoratum*** L.	Hypostomatic	31.07	23.50	1.32	16.24	19.56	5.14	3.75	1.37	<1	<1
9	***Galinsoga quadriradiata*** Ruiz & Pav.	Hypoamphistomatic	30.14	18.40	1.64	22.67	29.40	35.02	22.45	1.56	12.34	14.30
10	***Launea procumbens*** (Roxb.) Ramayya & Rajagopal	Hypoamphistomatic	23.24	10.13	2.29	20.68	26.24	24.20	10.67	2.27	13.93	16.19
11	***Parthenium hysterophorus*** L.	Epiamphistomatic	22.89	15.00	1.53	13.91	16.20	25.25	15.42	1.64	15.57	18.47
12	***Solidago canadensis*** L.	Hypoamphistomatic	30.94	23.18	1.33	14.36	16.80	33.09	26.47	1.25	8.27	9.02
13	***Sonchus wightianus*** DC.	Hypoamphistomatic	16.58	7.61	2.18	21.72	27.98	19.91	10.00	1.99	5.20	5.49
14	***Spilanthes calva*** DC.	Epiamphistomatic	19.42	11.38	1.71	20.05	25.11	18.33	11.17	1.64	21.03	26.69

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